WORKED SOLUTION

STEP II 2000, Question 11

Mechanics • Elastic Strings and SHM

STEP II 2000, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/00-s2-q11

The key idea

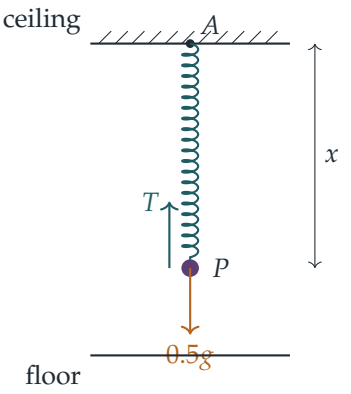
Energy settles both qualitative claims quickly: compare the elastic energy stored at release with the gravitational energy needed to reach the natural length and the ceiling. For the timing, split the motion into two phases: while the string is taut the particle moves in simple harmonic motion about its equilibrium position, and once the string is slack it moves freely under gravity. The trip back is the trip up run in reverse, so we double the time taken to go up.

Solution

We model P as a particle on a light elastic string obeying Hooke's law, with no air resistance. Let x be the distance of P below A, measured downwards, in metres. The string is taut when $x > 1.5$, and then its extension is $x - 1.5$ and its tension is

$$T = \frac{\lambda}{l}(x - 1.5) = \frac{5g}{1.5}(x - 1.5) = \frac{10g}{3}(x - 1.5).$$

The room is 2.5 m high, so P starts at $x = 2$, with extension 0.5 m.



The string becomes slack

At release, the elastic energy stored is

$$\frac{\lambda e^2}{2l} = \frac{5g\,(0.5)^2}{2(1.5)} = \frac{5g}{12}\ \text{J}.$$

For the string to reach its natural length, P must rise 0.5 m, gaining potential energy $0.5g \times 0.5 = \frac{g}{4}$ J. Since $\frac{5g}{12} > \frac{g}{4}$, P arrives at $x = 1.5$ still moving upwards, with kinetic energy

$$\tfrac{1}{2}(0.5)v^2 = \frac{5g}{12} - \frac{g}{4} = \frac{g}{6}, \qquad \text{so} \qquad v^2 = \frac{2g}{3}.$$

It therefore passes above the point where the string has its natural length, and the string becomes slack.

P does not reach the ceiling

To reach the ceiling, P would have to rise 2 m from its starting point, gaining $0.5g \times 2 = g$ J of potential energy. All the energy available is the $\frac{5g}{12}$ J stored in the string at release (once the string is slack it stores nothing), and $\frac{5g}{12} < g$. So P cannot reach the ceiling.

More precisely, after the string goes slack P is a projectile moving vertically with speed v, so it rises a further $\dfrac{v^2}{2g} = \dfrac{1}{3}$ m. Its highest point is at $x = 1.5 - \frac{1}{3} = \frac{7}{6}$, still $\frac{7}{6}$ m below the ceiling.

Simple harmonic motion while taut

For $x > 1.5$, Newton's second law (downwards positive) gives

$$0.5\,\ddot{x} = 0.5g - \frac{10g}{3}(x - 1.5).$$

Equilibrium is where the right-hand side vanishes: $x - 1.5 = 0.15$, so $x = 1.65$. Put $y = x - 1.65$, the displacement below equilibrium. Then

$$\ddot{y} = 2g - \frac{20g}{3}(y + 0.15) = -\frac{20g}{3}\,y,$$

which is simple harmonic motion about $y = 0$ with $\omega^2 = \dfrac{20g}{3}$. This holds whenever the string is in tension, as required.

The time to return

P starts at rest at $y = 2 - 1.65 = 0.35$, so while the string is taut

$$y = 0.35\cos\omega t.$$

The string goes slack when $x = 1.5$, that is $y = -0.15$. So the first time this happens, t_1, satisfies

$$\cos\omega t_1 = -\frac{0.15}{0.35} = -\frac{3}{7}.$$

Ask yourself Which solution of $\cos\omega t_1 = -\frac{3}{7}$ do we want, and how do we write it using arccos?

As t increases from 0, ωt increases from 0, and the first time $\cos\omega t$ reaches $-\frac{3}{7}$ is when ωt is in $(\frac{\pi}{2}, \pi)$. Since $\arccos\frac{3}{7} \in (0, \frac{\pi}{2})$ and $\cos(\pi - \theta) = -\cos\theta$, that value is $\omega t_1 = \pi - \arccos\frac{3}{7}$. Hence

$$t_1 = \frac{1}{\omega}\left(\pi - \arccos\tfrac{3}{7}\right) = \left(\frac{3}{20g}\right)^{1/2}\left(\pi - \arccos\tfrac{3}{7}\right).$$

(As a check, the speed then is $\omega\sqrt{0.35^2 - 0.15^2} = \sqrt{\frac{20g}{3} \times 0.1} = \sqrt{\frac{2g}{3}}$, matching the energy calculation.)

While the string is slack, P moves under gravity alone. It leaves $x = 1.5$ moving up with speed $v = \sqrt{2g/3}$, and it comes back to $x = 1.5$ moving down with the same speed, after time

$$t_2 = \frac{2v}{g} = \frac{2}{g}\sqrt{\frac{2g}{3}} = \left(\frac{8}{3g}\right)^{1/2}.$$

The motion under a force that depends only on position is reversible. Having re-entered the taut region at $x = 1.5$ with the same speed as it left, P retraces the SHM phase in reverse, taking another t_1 to come to rest at $y = 0.35$, its original position. The total time is

$$\begin{aligned} 2t_1 + t_2 &= 2\left(\frac{3}{20g}\right)^{1/2}\left(\pi - \arccos\tfrac{3}{7}\right) + \left(\frac{8}{3g}\right)^{1/2} \\ &= \left(\frac{8}{3g}\right)^{1/2} + \left(\frac{3}{5g}\right)^{1/2}\left(\pi - \arccos\tfrac{3}{7}\right), \end{aligned}$$

using $2\sqrt{3/(20g)} = \sqrt{12/(20g)} = \sqrt{3/(5g)}$. This is the required result.

Checks

- The two routes to the speed at the natural length (energy, and $v^2 = \omega^2(a^2 - y^2)$ for SHM) both give $v^2 = \frac{2g}{3}$.
- With $g = 9.8$, the formula gives about 1.0199 s. Stepping the equation of motion numerically (with the tension switched off when $x < 1.5$) gives the same time to return to rest at the start, and a highest point $\frac{7}{6}$ m below the ceiling.
- Without the slack phase, the SHM period would be $2\pi/\omega \approx 0.77$ s. Our $2t_1 \approx 0.50$ s is less than this, as it should be, since only part of the oscillation is completed.

Common slips

- Writing the SHM about the natural length instead of the equilibrium position. Gravity shifts the centre to an extension of 0.15 m.

- Taking $\omega t_1 = \arccos(-\frac{3}{7})$ and then not recognising that this is $\pi - \arccos\frac{3}{7}$.
- Forgetting the return journey: the time asked for includes going up *and* coming back, so both phases are doubled.

Beyond STEP

This is a piecewise-defined, or "non-smooth", dynamical system: different equations of motion hold in different regions, and solutions are pieced together at the boundaries by matching position and velocity. The same idea governs bouncing balls, impacts and switching circuits, and the reversibility argument we used is a first glimpse of time-reversal symmetry in conservative systems. For more, see Kleppner and Kolenkow, *An Introduction to Mechanics*.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.